Classical Mechanics

Lecture 8: Poisson Brackets, Generators, and Angular Momentum

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Chapter 1

Poisson Brackets, Generators, and Angular Momentum

Overview. The Poisson bracket is more than a compact way to rewrite Hamilton's equations. It is an algebraic operation that generates transformations. Bracketing with the Hamiltonian generates time evolution; bracketing with momentum generates translations; bracketing with angular momentum generates rotations. In this language, symmetry and conservation become the two readings of one vanishing bracket. We shall develop that idea from the harmonic oscillator and then use the angular-momentum algebra to obtain gyroscopic precession without introducing Euler angles.

1.1 A Function Along a Hamiltonian Trajectory

Let

$$F = F(q_1, \dots, q_n, p_1, \dots, p_n)$$

be a sufficiently differentiable function on phase space. It can be viewed in two ways. First, it assigns a number to every point of phase space. Second, once a Hamiltonian trajectory has been chosen, it becomes a function of time because the point $(q(t), p(t))$ moves. Along that trajectory,

$$\frac{dF}{dt} = \sum_i \left(\frac{\partial F}{\partial q_i}\dot{q}_i + \frac{\partial F}{\partial p_i}\dot{p}_i \right). \tag{1.1}$$

Hamilton's equations,

$$\dot{q}_i = \frac{\partial H}{\partial p_i}, \qquad \dot{p}_i = -\frac{\partial H}{\partial q_i}, \tag{1.2}$$

turn this into

$$\frac{dF}{dt} = \sum_i \left(\frac{\partial F}{\partial q_i}\frac{\partial H}{\partial p_i} - \frac{\partial F}{\partial p_i}\frac{\partial H}{\partial q_i} \right). \tag{1.3}$$

The combination on the right occurs throughout classical mechanics, even when neither of its entries is the Hamiltonian. This motivates the definition

$$\boxed{\{F, G\} = \sum_i \left(\frac{\partial F}{\partial q_i}\frac{\partial G}{\partial p_i} - \frac{\partial F}{\partial p_i}\frac{\partial G}{\partial q_i} \right).} \tag{1.4}$$

Equation (1.3) is therefore

$$\boxed{\dot{F} = \{F, H\}.} \tag{1.5}$$

Question from the audience. If the second entry is changed from H to an arbitrary function G, is the bracket still the time derivative of F? [a]

Response. No. The audience catches an important point while the notation is being generalized. Equation (1.4) defines $\{F, G\}$ for any two phase-space functions, but $\dot{F} = \{F, G\}$ holds only when G is the Hamiltonian. A general G will generate a different infinitesimal transformation.

[a] Lecture timestamp: 00:05:45–00:06:15.

It is useful to write a small change in time as

$$\delta_t F = \epsilon \{F, H\}. \tag{1.6}$$

This form anticipates what comes later. Time evolution is one member of a larger family of transformations generated by Poisson brackets.

1.2 The Algebra of Brackets

The bracket can now be used as a calculus of its own. Its rules follow from ordinary differentiation:

$$\{A, B\} = -\{B, A\}, \qquad \text{antisymmetry,} \tag{1.7}$$

$$\{A + B, C\} = \{A, C\} + \{B, C\}, \qquad \text{additivity,} \tag{1.8}$$

$$\{\lambda A, B\} = \lambda \{A, B\}, \qquad \lambda \text{ constant,} \tag{1.9}$$

$$\{AB, C\} = A\{B, C\} + \{A, C\}B, \qquad \text{Leibniz rule.} \tag{1.10}$$

Classical phase-space functions multiply commutatively, so the placement of the ordinary factors A and B in the last line is immaterial. That will no longer be true for operators in quantum mechanics. Antisymmetry also implies

$$\{A, A\} = 0. \tag{1.11}$$

For canonical coordinates the elementary brackets are

$$\boxed{\{q_i, q_j\} = 0, \qquad \{p_i, p_j\} = 0, \qquad \{q_i, p_j\} = \delta_{ij}.} \tag{1.12}$$

Any function only of the q_i has zero bracket with any other function only of the q_i; the analogous statement holds for functions only of the p_i. A nonzero bracket requires a canonical coordinate to meet its conjugate momentum. The Kronecker delta in Eq. (1.12) records that only matching indices meet.

1.2.1 The harmonic oscillator as a test

Consider

$$H = \frac{p^2}{2m} + \frac{kq^2}{2}. \tag{1.13}$$

We now deliberately use only the bracket rules and Eq. (1.5). First,

$$\dot q = \{q, H\} \tag{1.14}$$
$$= \frac{1}{2m}\{q, p^2\} \tag{1.15}$$
$$= \frac{1}{2m}\left(p\{q, p\} + \{q, p\}p\right) \tag{1.16}$$
$$= \frac{p}{m}. \tag{1.17}$$

The potential term disappears because $\{q, q^2\} = 0$. Similarly,

$$\dot p = \{p, H\} \tag{1.18}$$
$$= \frac{k}{2}\{p, q^2\} \tag{1.19}$$
$$= \frac{k}{2}\left(q\{p, q\} + \{p, q\}q\right) \tag{1.20}$$
$$= -kq. \tag{1.21}$$

The minus sign is precisely the antisymmetry $\{p, q\} = -\{q, p\} = -1$. Combining the two first-order equations gives

$$m\ddot q = -kq. \tag{1.22}$$

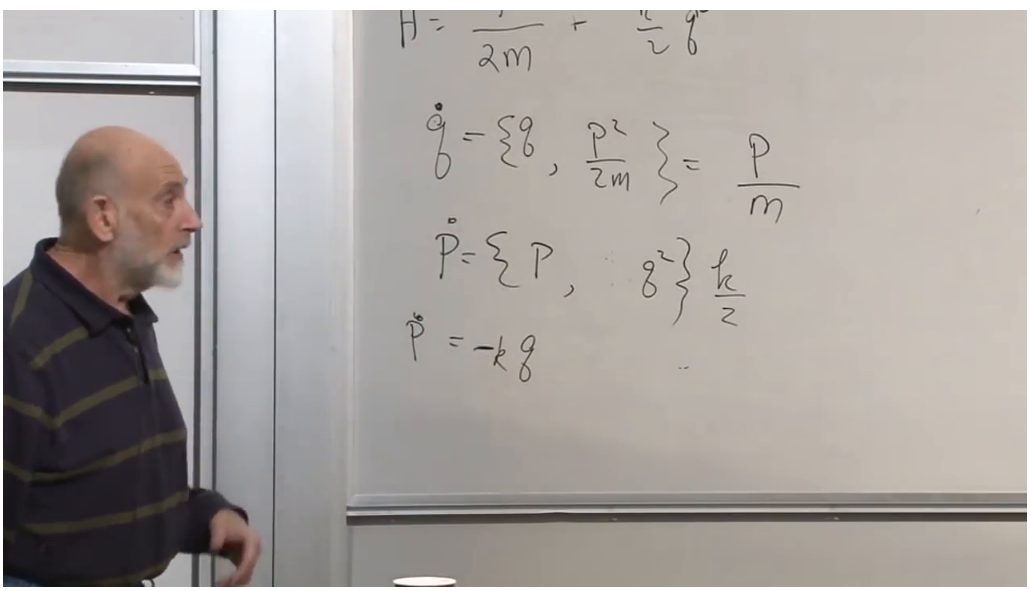

Figure 1.1: The oscillator Hamiltonian and the two bracket calculations yielding $\dot q = p/m$ and $\dot p = -kq$.

Lecture frame: 00:23:20.

During the calculation the class looks for a verb meaning *take a Poisson bracket* and settles, provisionally, on *Poissonate*. The joke marks a real change of viewpoint: once the algebra is known, one can work with it directly instead of re-deriving every result from the definition.

Question from the audience. Is this mathematical structure a group, or an algebra? Does it have inverses? [a]

Response. It is not a group: the bracket operation does not provide the required identity and inverses. The phase-space functions form a Poisson algebra: they have ordinary commutative multiplication and addition, together with an antisymmetric bracket obeying the Leibniz rule. A further identity, the Jacobi identity, completes the usual abstract definition, although it is not developed in this lecture.

[a] Lecture timestamp: 00:23:48–00:24:16.

Question from the audience. What is the connection between Poisson brackets and quantum commutators? [a]

Response. The lecture gives only a preview. In the passage to quantum mechanics, the classical bracket is replaced schematically by

$$\{A, B\}_{\mathrm{PB}} \longleftrightarrow \frac{1}{i\hbar}[\widehat{A}, \widehat{B}].$$

This correspondence requires qualifications about operator ordering and quantization; it is not being used as a theorem here.

[a] Lecture timestamp: 00:24:18–00:25:00.

For the oscillator, the bracket has merely reconstructed Hamilton's equations. Its real economy appears when the generator is a conserved quantity other than H.

1.3 Angular Momentum Recovered from Rotational Symmetry

Recall the infinitesimal form of Noether's construction used earlier in the course. If a continuous symmetry changes the coordinates by

$$\delta q_i = \epsilon F_i(q), \tag{1.23}$$

then, in the setting considered here, its conserved quantity is

$$G = \sum_i p_i F_i(q). \tag{1.24}$$

The overall factor ϵ may be omitted: multiplying a conserved quantity by a fixed number does not change the conservation law.

Take one particle and rotate it infinitesimally about the z-axis. With the orientation used in the lecture,

$$\delta x = -\epsilon y, \qquad \delta y = \epsilon x, \qquad \delta z = 0. \tag{1.25}$$

Equation (1.24) gives

$$L_z = -p_x y + p_y x = x p_y - y p_x. \tag{1.26}$$

Cycling x, y, and z,

$$\boxed{L_x = y p_z - z p_y, \qquad L_y = z p_x - x p_z, \qquad L_z = x p_y - y p_x.} \tag{1.27}$$

These are the components of

$$\boldsymbol{L} = \boldsymbol{r} \times \boldsymbol{p}. \tag{1.28}$$

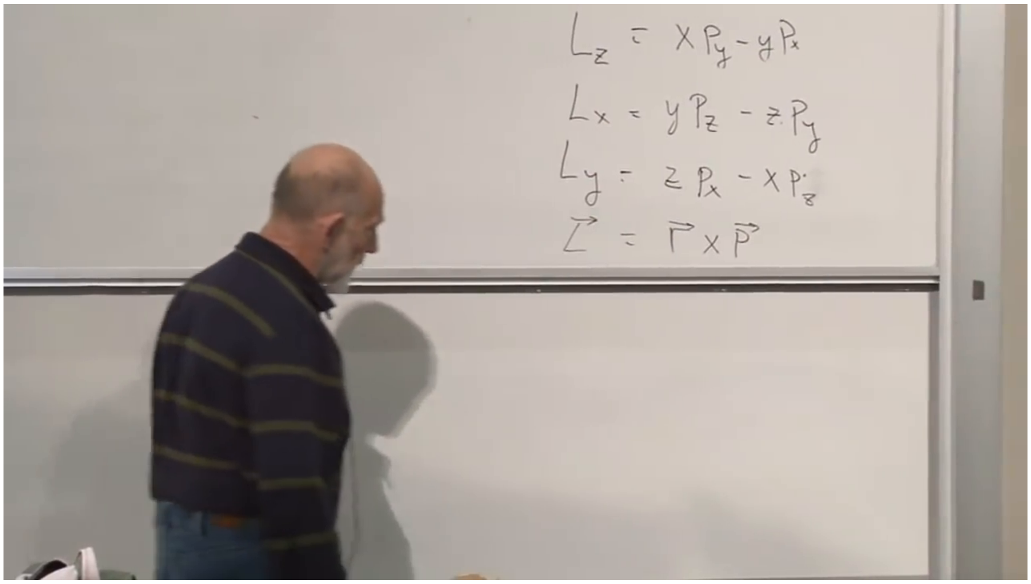

Figure 1.2: The three Cartesian components of angular momentum and their compact cross-product form.

Lecture frame: 00:34:00.

Rotation about an arbitrary unit vector $\boldsymbol{n}$ is generated by

$$L_{\boldsymbol{n}} = \boldsymbol{n} \cdot \boldsymbol{L}. \tag{1.29}$$

For an isolated rotationally invariant system, all three components are conserved. If only rotations around one axis remain a symmetry, only the component along that axis is guaranteed to be conserved.

1.4 Angular Momentum Generates Rotations

Instead of immediately solving a rotational dynamics problem, let us explore what L_z does under the bracket. Using Eq. (1.26),

$$\{x, L_z\} = \{x, xp_y - yp_x\} = -y\{x, p_x\} = -y, \tag{1.30}$$

$$\{y, L_z\} = \{y, xp_y - yp_x\} = x\{y, p_y\} = x, \tag{1.31}$$

$$\{z, L_z\} = 0. \tag{1.32}$$

These are exactly the coefficients in Eq. (1.25).

Momentum is also a vector, and the same calculation gives

$$\{p_x, L_z\} = -p_y, \qquad \{p_y, L_z\} = p_x, \qquad \{p_z, L_z\} = 0. \tag{1.33}$$

The general Hamiltonian statement is therefore

$$\boxed{\delta_G A = \epsilon\{A, G\}.} \tag{1.34}$$

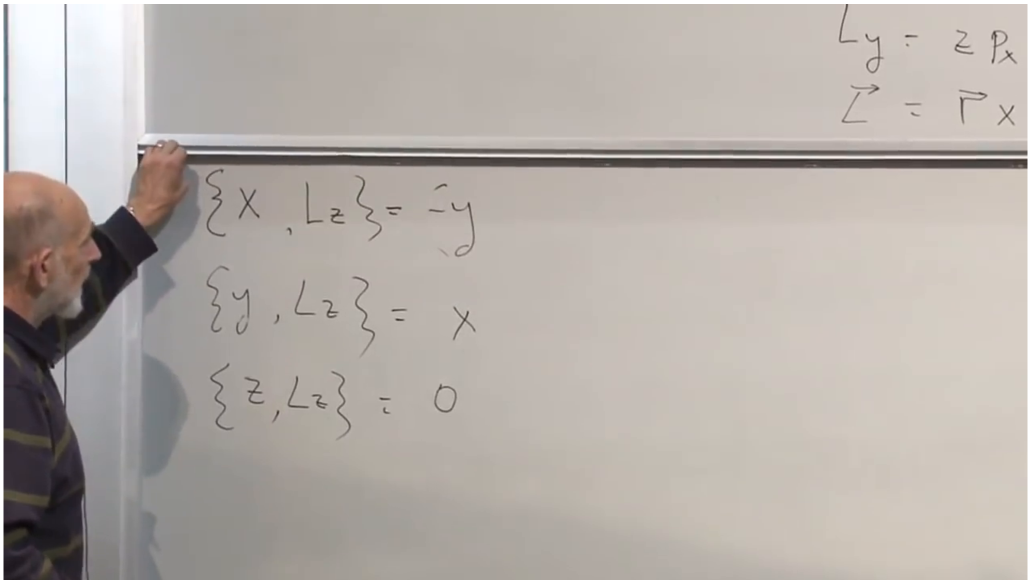

Figure 1.3: The coordinate brackets with L_z: the bracket reproduces the infinitesimal rotation of (x, y, z).

Lecture frame: 00:38:41.

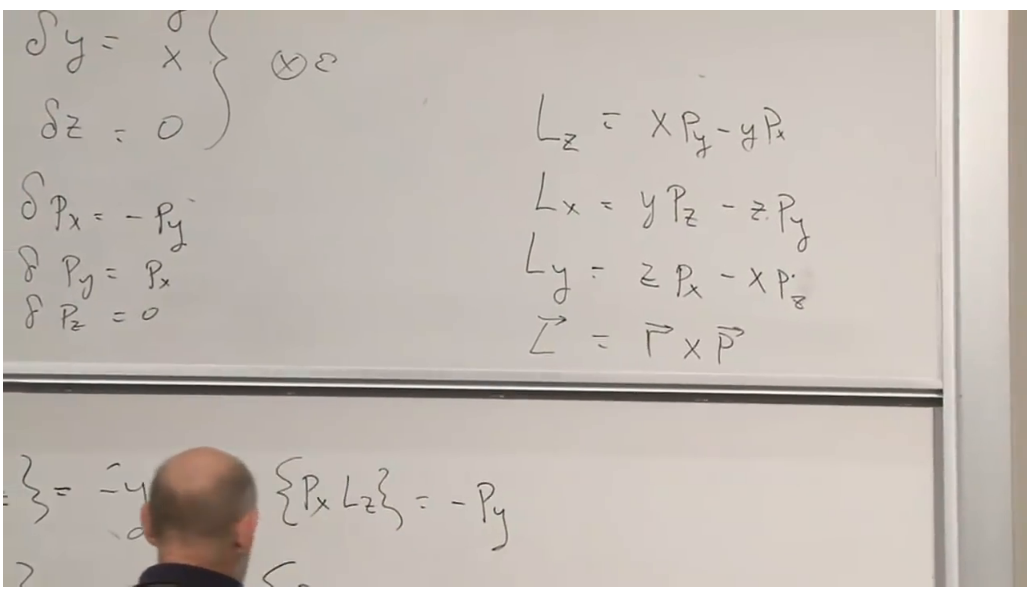

Figure 1.4: Position and momentum transform with the same vector pattern under the rotation generated by L_z.

Lecture frame: 00:42:08.

Every phase-space function G defines such an infinitesimal canonical transformation. If $G = L_z$, it is a rotation. If $G = H$, it is a small step in time.

Question from the audience. The lecture sometimes speaks of rotating the system and sometimes of rotating coordinates. Which convention is used in the bracket equations? [a]

Response. Equation (1.25) fixes the convention: it is the infinitesimal active rotation $(x, y) \mapsto (x - \epsilon y, y + \epsilon x)$. A passive rotation of the axes carries the opposite sign. Once the convention is fixed, the bracket formulas are unambiguous.

[a] Lecture timestamp: 00:28:22–00:29:35.

1.5 Momentum Generates Translations

A translation in one coordinate has

$$\delta q = \epsilon. \tag{1.35}$$

Here $F(q) = 1$ in Eq. (1.23), so the corresponding conserved quantity is p. The generator claim predicts

$$\{f(q), p\} = \frac{df}{dq}. \tag{1.36}$$

The lecture proves this first for powers by induction. The base case is $\{q, p\} = 1$. Suppose

$$\{q^{n-1}, p\} = (n-1)q^{n-2}. \tag{1.37}$$

Then the product rule gives

$$\begin{aligned}
\{q^n, p\} &= \{q\, q^{n-1}, p\} && (1.38)\\
&= q\{q^{n-1}, p\} + \{q, p\}q^{n-1} && (1.39)\\
&= (n-1)q^{n-1} + q^{n-1} && (1.40)\\
&= nq^{n-1}. && (1.41)
\end{aligned}$$

Linearity extends the result to polynomials and to convergent power-series representations. More directly, the definition of the bracket gives Eq. (1.36) for every differentiable $f(q)$:

$$\{f(q), p\} = \frac{\partial f}{\partial q}\frac{\partial p}{\partial p} = f'(q). \tag{1.42}$$

We have now seen three versions of one idea:

Generator	Transformation	Infinitesimal action
H	time translation	$\delta_t A = \epsilon\{A, H\}$
p_i	spatial translation	$\delta A = \epsilon\{A, p_i\}$
L_i	rotation	$\delta A = \epsilon\{A, L_i\}$

The conserved quantity associated with a continuous symmetry is also the generator of that symmetry.

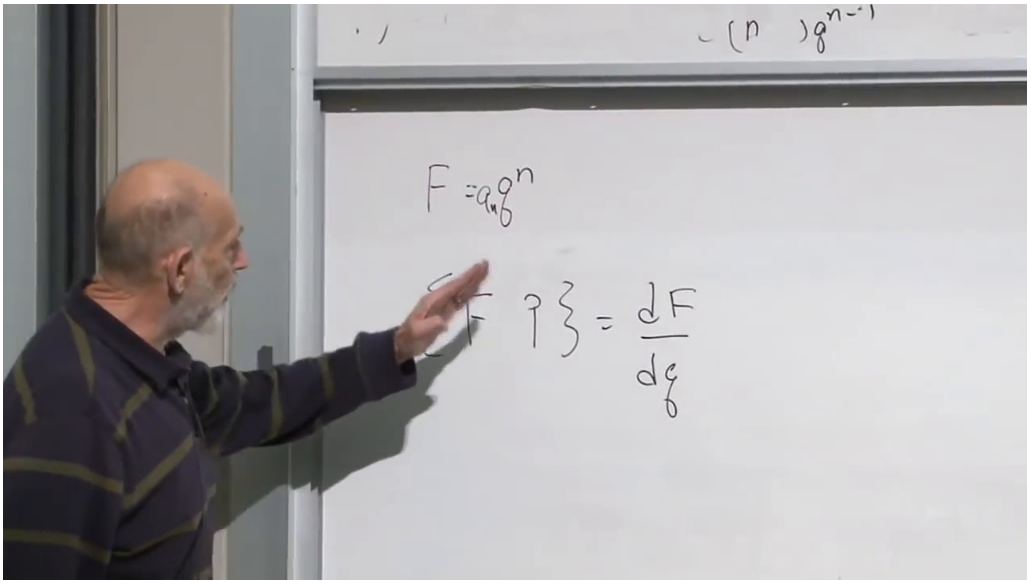

Figure 1.5: The translation result after the induction argument: $\{F(q), p\} = dF/dq$.
Lecture frame: 00:53:18.

1.6 One Bracket, Two Readings

Let $G(q, p)$ be a proposed generator. If it is conserved and has no explicit time dependence, then

$$\dot{G} = \{G, H\} = 0. \tag{1.43}$$

By antisymmetry,

$$\{H, G\} = 0. \tag{1.44}$$

The equations are mathematically equivalent, but their physical readings are different:

- $\{G, H\} = 0$ says that G does not change under time evolution generated by H.
- $\{H, G\} = 0$ says that H does not change under the transformation generated by G.

The first is a conservation law; the second is a symmetry of the Hamiltonian. Noether's connection has been compressed into one zero.

Question from the audience. If a conserved quantity is known, can the associated symmetry always be recovered? [a]

Response. Within this Hamiltonian setting, yes: bracket every phase-space variable with G. The collection $\delta_G A = \epsilon\{A, G\}$ tells how the variables transform. For example, the brackets with L_z reveal a rotation even if that interpretation was not known in advance. Global and boundary subtleties can arise in more advanced settings, but the local infinitesimal statement is the one used here.

[a] Lecture timestamp: 01:01:42–01:02:15.

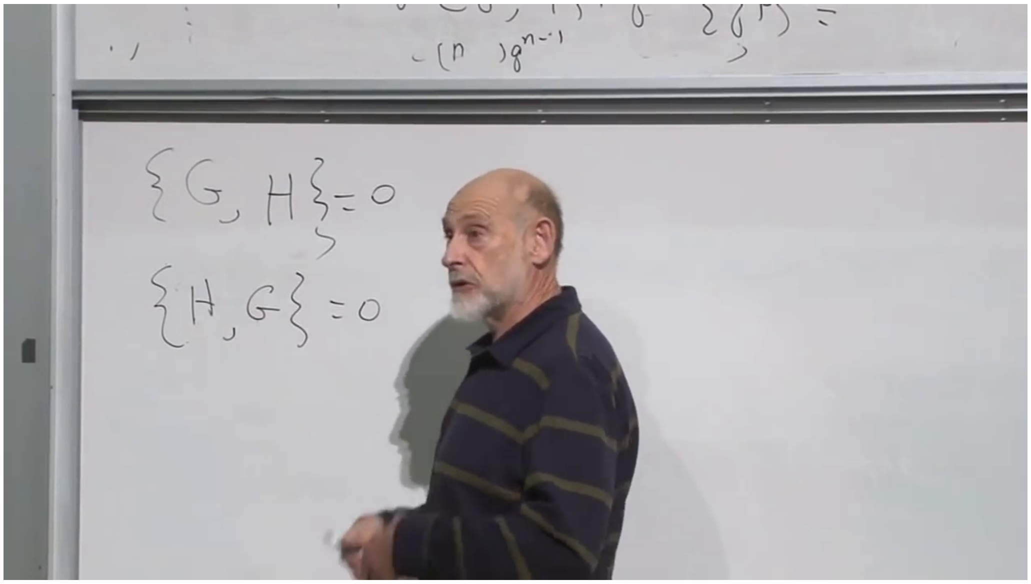

Figure 1.6: The two orders of the same vanishing bracket: conservation of G and invariance of H under the G-generated transformation.

Lecture frame: 01:03:05.

Question from the audience. Do those bracket equations tell us the physical motion responsible for the conservation law? [a]

Response. No. They identify the symmetry transformation, not the actual trajectory. Conservation of L_z indicates rotational symmetry; it does not say that the particle moves in a circle. Actual time evolution is generated by H. A bracket with another generator describes another direction of transformation in phase space.

[a] Lecture timestamp: 00:56:01–00:58:13.

Question from the audience. Is there a general transformation statement that does not already assume conservation? [a]

Response. Yes. Equation (1.34) defines an infinitesimal transformation for every function G, whether or not it is conserved. It becomes a symmetry only when the Hamiltonian is invariant: $\{H, G\} = 0$. The same condition, read in the opposite order, then says that G is conserved.

[a] Lecture timestamp: 01:02:40–01:04:45.

1.7 The Angular-Momentum Algebra

The next calculation lets us stop dragging the particle coordinates through every rotational problem. Begin with

$$L_x = yp_z - zp_y, \qquad L_z = xp_y - yp_x.$$

In $\{L_x, L_z\}$, most cross terms vanish because their indices do not form canonical pairs. The surviving contractions give

$$\{L_x, L_z\} = \{yp_z - zp_y, xp_y - yp_x\} \tag{1.45}$$
$$= xp_z - zp_x \tag{1.46}$$
$$= -L_y. \tag{1.47}$$

Reversing the order removes the minus sign. Cycling the coordinates produces

$$\boxed{\{L_x, L_y\} = L_z, \qquad \{L_y, L_z\} = L_x, \qquad \{L_z, L_x\} = L_y.} \tag{1.48}$$

In index notation,

$$\{L_i, L_j\} = \varepsilon_{ijk} L_k. \tag{1.49}$$

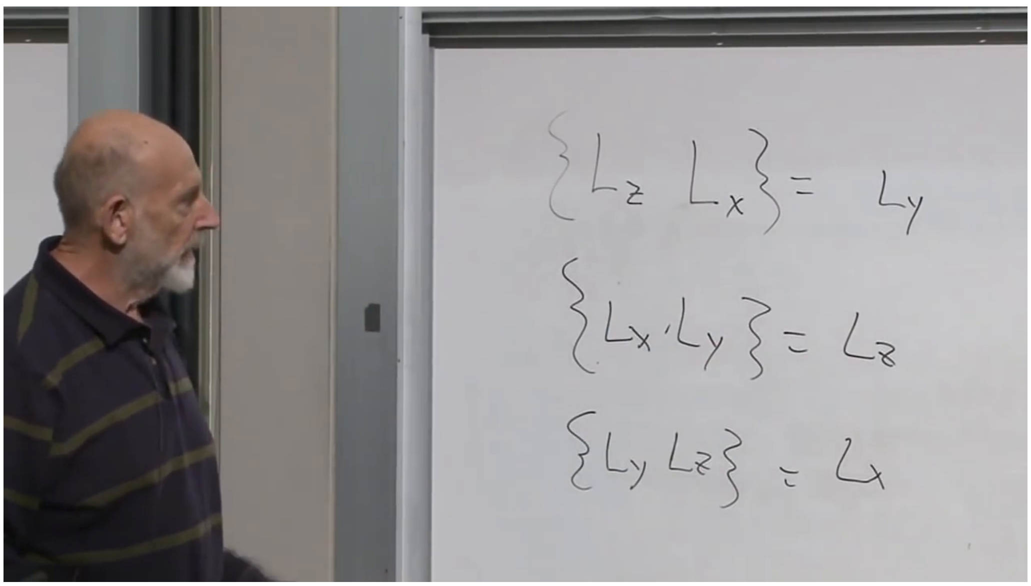

Figure 1.7: The cyclic Poisson-bracket algebra of the three angular-momentum components.
Lecture frame: 01:10:52.

The result could also have been anticipated. Since $\boldsymbol{L}$ is a vector, bracketing its x-component with L_z must give the infinitesimal change of that component under a z-rotation, namely $-L_y$. The explicit calculation confirms that geometric argument. These equations are also the classical precursors of the quantum angular-momentum commutation relations.

1.8 A Fast Gyroscope

The payoff is a gyroscope: a rapidly spinning flywheel whose center is a fixed distance R from a pivot. Let $\boldsymbol{r}$ point from the pivot to the wheel center. The flywheel spins so rapidly about its axle that its dominant angular momentum remains aligned with that direction. In the approximation used in the lecture,

$$\boldsymbol{L} = \ell\,\boldsymbol{r}, \qquad L_z = \ell z, \tag{1.50}$$

where $\ell = L/R$ is treated as constant. Smaller angular momenta associated with nutation or motion of the axle are neglected at first.

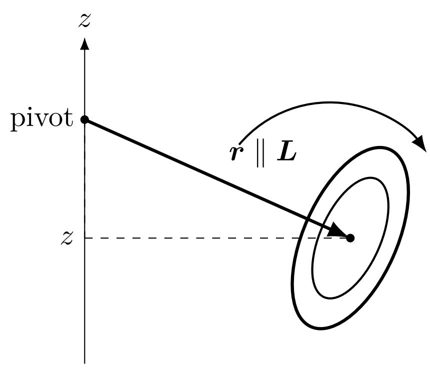

Figure 1.8: Fast-gyroscope approximation. The pivot-to-center distance is fixed, and the dominant spin angular momentum remains aligned with the axle and $\boldsymbol{r}$.

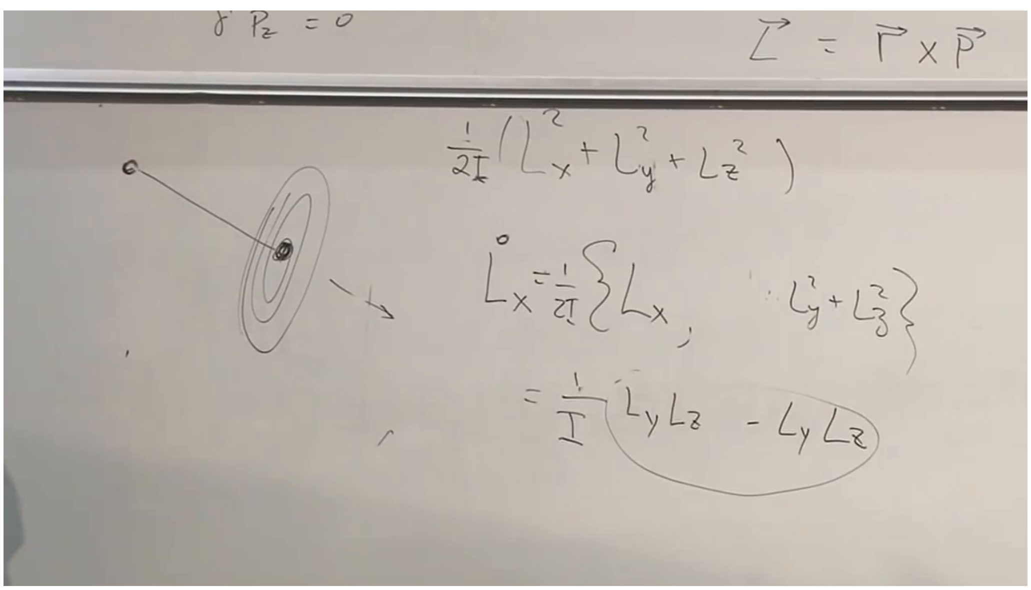

Figure 1.9: The board sketch of the flywheel together with $H_{\mathrm{rot}} = L^2/(2I)$ and the cancellation proving torque-free conservation of $\boldsymbol{L}$.

Lecture frame: 01:18:54.

1.8.1 No gravity: why every component stays fixed

For the simplified axially symmetric rotor,

$$H_{\mathrm{rot}} = \frac{1}{2I}\left(L_x^2 + L_y^2 + L_z^2\right) = \frac{L^2}{2I}. \tag{1.51}$$

The coefficient is $1/(2I)$, not $I/2$; that slip is made and corrected at the board. Using Eq. (1.48),

$$\dot{L}_x = \{L_x, H_{\mathrm{rot}}\} \tag{1.52}$$

$$= \frac{1}{2I}\left(2L_y\{L_x, L_y\} + 2L_z\{L_x, L_z\}\right) \tag{1.53}$$

$$= \frac{1}{I}\left(L_yL_z - L_zL_y\right) = 0. \tag{1.54}$$

Cyclically,

$$\dot{L}_x = \dot{L}_y = \dot{L}_z = 0. \tag{1.55}$$

The scalar L^2 has zero bracket with each component L_i; in the language of this algebra, it is a Casimir. Thus a torque-free gyroscope keeps both the magnitude and direction of its angular momentum.

Question from the audience. Is the moment of inertia just a number here? [a]

Response. For this reduced model, yes. It encodes the mass distribution of the wheel about its spin axis. A general rigid body has a moment-of-inertia tensor and a rotational Hamiltonian $\sum_i L_i^2/(2I_i)$ in principal axes. The lecture chooses the simpler fast, symmetric rotor because the aim is to display the Poisson-bracket method rather than develop full rigid-body dynamics.

[a] Lecture timestamp: 01:14:23–01:15:05.

1.8.2 Gravity and the sign of the potential

Take the vertical coordinate z positive upward and put the pivot at the origin. The gravitational potential energy of the wheel center is

$$V = Mgz. \tag{1.56}$$

Since $L_z = \ell z$,

$$V = cL_z, \qquad c = \frac{Mg}{\ell} = \frac{MgR}{L}. \tag{1.57}$$

Here $c > 0$ when $\boldsymbol{L}$ is chosen parallel to $\boldsymbol{r}$. If the flywheel spins in the opposite sense, the signed relation between $\boldsymbol{L}$ and $\boldsymbol{r}$ reverses and so does the corresponding signed precession frequency.

Question from the audience. Why may height be replaced by a component of angular momentum? Is the slanted line an axle or a position vector? [a]

Response. The slanted line is best labeled $\boldsymbol{r}$, the vector from the fixed pivot to the wheel center. A physical axle may lie along it, but the relation used in the calculation is the fast-spin approximation $\boldsymbol{L} = \ell\boldsymbol{r}$. Their z-components are therefore proportional. The live discussion changes labels several times before settling on this distinction.

[a] Lecture timestamp: 01:22:20–01:25:43.

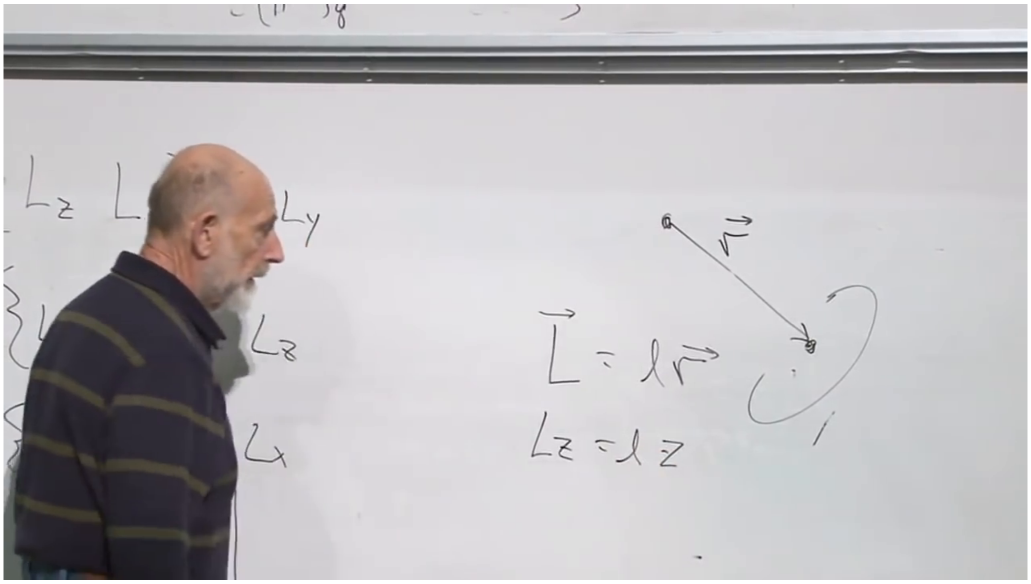

Figure 1.10: The fixed-length geometry behind $\boldsymbol{L} = \ell \boldsymbol{r}$ and $L_z = \ell z$.

Lecture frame: 01:25:38.

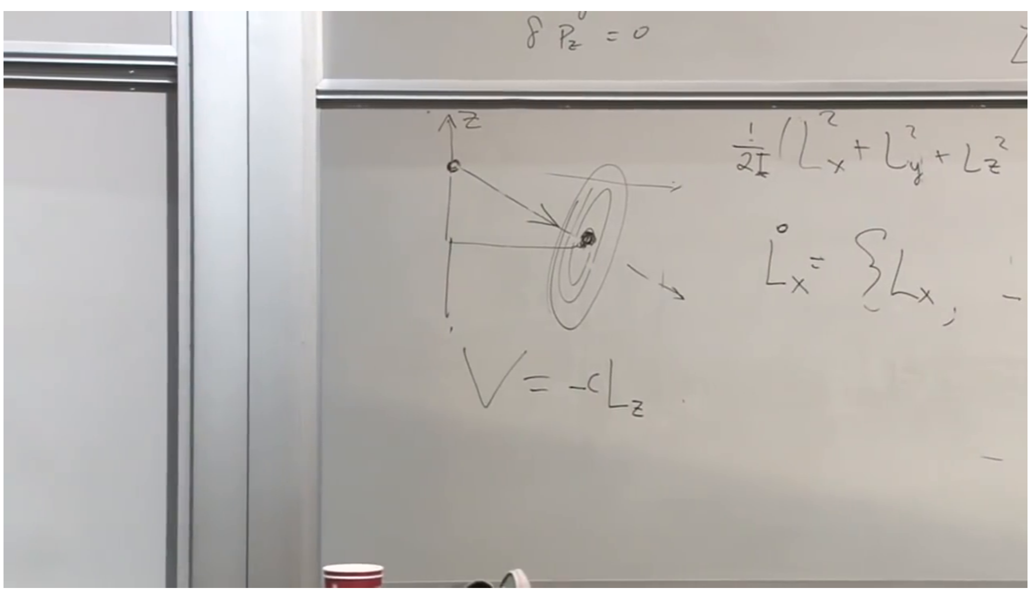

Figure 1.11: An intermediate board state shows $V = -cL_z$. The sign is corrected later after z is fixed as upward: with $\boldsymbol{L} = \ell \boldsymbol{r}$ and $\ell > 0$, the consistent result is $V = +cL_z$.

Lecture frame: 01:22:43.

Question from the audience. What assumptions make ℓ constant, and what fails if the wheel center can slide radially? [a]

Response. The dominant spin is assumed much larger than the angular momentum associated with the slow motion, and the pivot-to-center distance is fixed. Then the spin magnitude changes little and $\boldsymbol{L}$ rotates with $\boldsymbol{r}$. If the center could slide radially, z would no longer be proportional to L_z with one constant coefficient, so the reduced Hamiltonian $V = cL_z$ would not describe the motion.

[a] Lecture timestamp: 01:25:43–01:26:38, 01:34:22–01:35:00.

Question from the audience. Why does the board first show a minus sign and later a plus sign? Does the spin direction matter? [a]

Response. The initial sign is a live-board error mixed with an unsettled axis convention. Once z is defined upward, $V = Mgz$ increases with height. If $\boldsymbol{L} = \ell\boldsymbol{r}$ with $\ell > 0$, this gives $V = +cL_z$. Reversing the spin reverses the signed proportionality and the direction of precession, but it does not restore an arbitrary sign choice.

[a] Lecture timestamp: 01:27:13–01:28:15, 01:33:09–01:33:21.

1.8.3 Precession from three short equations

The full reduced Hamiltonian is

$$H = \frac{1}{2I}\left(L_x^2 + L_y^2 + L_z^2\right) + cL_z. \tag{1.58}$$

We have already shown that L^2 has zero bracket with every L_i. Only the linear term contributes:

$$\dot{L}_z = \{L_z, H\} = 0, \tag{1.59}$$

$$\dot{L}_x = \{L_x, H\} = c\{L_x, L_z\} = -cL_y, \tag{1.60}$$

$$\dot{L}_y = \{L_y, H\} = c\{L_y, L_z\} = cL_x. \tag{1.61}$$

The vertical component and the horizontal magnitude are constant:

$$\frac{d}{dt}(L_x^2 + L_y^2) = 2L_x(-cL_y) + 2L_y(cL_x) = 0. \tag{1.62}$$

Combining the horizontal components,

$$L_+ = L_x + iL_y, \qquad \dot{L}_+ = icL_+, \tag{1.63}$$

so

$$L_+(t) = L_+(0)e^{ict}. \tag{1.64}$$

The angular-momentum vector therefore precesses around the vertical axis with angular speed $|c| = MgR/L$. Its tilt remains fixed in this idealized steady motion.

Question from the audience. Was the rotational-energy term neglected when the precession equations were written? [a]

Response. No. It belongs in the full Hamiltonian as in Eq. (1.58). It simply contributes zero to $\dot{L}_i$ because $\{L_i, L^2\} = 0$, a cancellation already proved in the torque-free calculation. The student question makes this implicit step explicit.

[a] Lecture timestamp: 01:35:01–01:35:49.

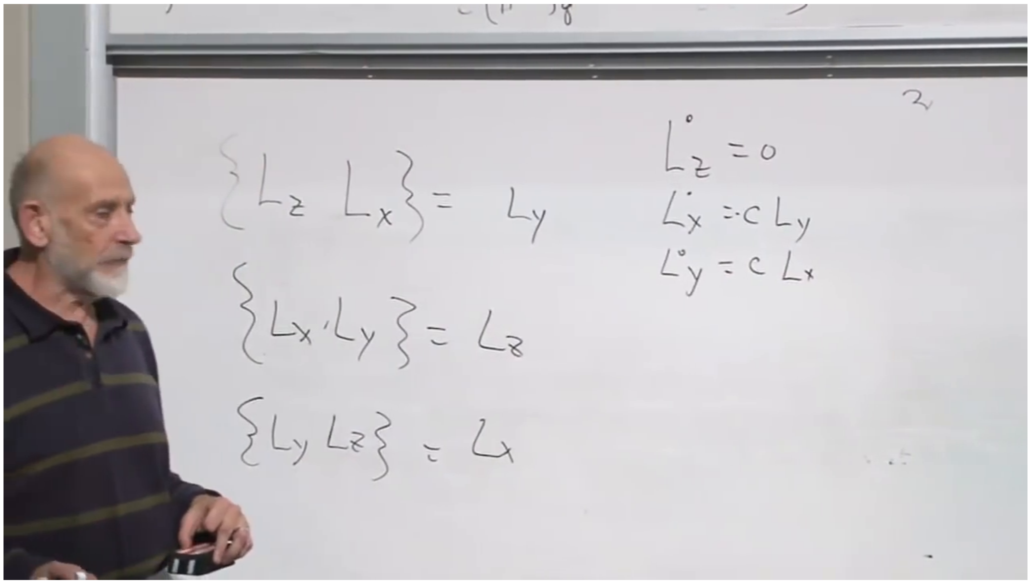

Figure 1.12: The final component equations: L_z is constant while L_x and L_y rotate into one another. *Lecture frame: 01:30:35.*

Question from the audience. Is uniform precession an equilibrium? What happens after a small kick? [a]

Response. It is better called a steady or stationary motion than a static equilibrium. A small kick generally adds nutation, an oscillation of the tilt angle. For a very rapidly spinning wheel that oscillation is small, which is why the reduced treatment keeps only nearly uniform precession.

[a]Lecture timestamp: 01:32:38–01:33:08.

Question from the audience. Does precession require a support or a real axle? [a]

Response. This mechanical model assumes a fixed pivot-to-center distance, so some constraint supplies the support. The deeper algebraic result is broader: whenever an angular momentum has a Hamiltonian term linear in one component, the same bracket algebra produces precession. The microscopic source of the angular momentum need not be a literal wheel on an axle.

[a]Lecture timestamp: 01:31:49–01:32:38, 01:33:49–01:35:00.

Question from the audience. Is this the same precession as electron or nuclear spin in a magnetic field, or as a planet's orbit precessing? [a]

Response. A magnetic moment coupled to a magnetic field has a Hamiltonian linear in a spin component and therefore obeys the same angular-momentum algebra. Nuclear-spin precession underlies nuclear magnetic resonance. The precession of a planetary orbit's perihelion is a different phenomenon that happens to share the word *precession*; it is not the same calculation.

[a]Lecture timestamp: 01:35:50–01:37:00.

Question from the audience. Is the side-mounted gyroscope essentially the same object as a spinning top? [a]

Response. Yes. The distinction is mostly geometric language: when the spinning axis extends sideways from its support one usually says gyroscope; when it is nearly vertical one says top. The angular-momentum and torque structure is the same.

[a] Lecture timestamp: 01:37:02–01:37:23.

1.9 What the Calculation Accomplished

No Euler angles were introduced, and no coordinates for every piece of the wheel were followed. Once

$$\{L_i, L_j\} = \varepsilon_{ijk} L_k$$

was known, the equations for the variables of interest followed directly. That is the intended demonstration of power. The oscillator showed that Poisson brackets can reproduce familiar dynamics. Angular momentum showed that they do much more:

- a conserved quantity generates its associated infinitesimal symmetry;
- conservation of G and invariance of H are the two readings of $\{G, H\} = 0$;
- the algebra of observables can close on itself, allowing dynamics to be solved without returning to microscopic coordinates;
- a Hamiltonian linear in one angular-momentum component produces precession around that axis.

The lecture stops before the full Euler equations for a general rigid body. Its narrower conclusion is already complete: the right generators and their Poisson algebra expose both symmetry and motion with remarkable economy.